

Sheet 3 — Matrices

Numbering continues from Sheet 2.

- 39.** Given the matrix $A = \begin{bmatrix} 3 & 4 & 5 \\ 2 & 1 & 3 \\ 0 & -1 & 2 \\ -4 & 5 & -8 \end{bmatrix}$: a) state the size of this matrix; b) identify in the matrix the elements a_{11} , a_{22} , a_{12} , a_{32} , a_{43} .

- 40.** State the special name of each of the following matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}; \quad B = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}; \quad C = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 4 & 5 \\ 5 & 2 & 1 \end{bmatrix}; \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad E = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix};$$

$$F = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}; \quad G = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 5 & -4 \end{bmatrix}; \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}; \quad M = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix};$$

$$J = \begin{bmatrix} 2+i & 3 & 0 \\ 0 & 4i & 3i \\ 0 & 0 & 7 \end{bmatrix}; \quad K = \begin{bmatrix} x & 0 & 0 \\ 0 & x^2 & 0 \\ 0 & 0 & e^x \end{bmatrix}; \quad L = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

- 41.** Let U be the real vector space of matrices of size 2×3 and let V be the subset of U of the elements of the form $\begin{bmatrix} a & b & a+b \\ 0 & 2b & 3a-b \end{bmatrix}$, $a, b \in \mathbb{R}$.

- a) Let A and B be the elements of V obtained for $a = b = 1$ and $a = 2 \wedge b = -1$, respectively. Solve the matrix equation $2A + X = B$.
- b) Show that V is a subspace of U and find its dimension.

- 42.** Given the matrices $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, compute: a) $A \times B + C$ b) $2B \times 3A + I_3$ c) C^2

- 43.** Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $D = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

- a) Compute $3A - 2B$.
- b) Solve the equation $2A - 3X = B$.
- c) Compute AC , AD , $A(CD)$, AB , BA , ABC , $ACD^T B$, $C^T BC$.
- d) Find all the matrices Y that commute with A , i.e. such that $AY = YA$.

- 44.** Consider the matrices $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & 3 \end{bmatrix}$. Which of the following products are possible? Compute them, together with their traces: AB , AB^T , $A^T B$, BA , BA^T .

- 45.** Compute the rank of the following matrices: $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & -1 \\ 2 & 1 \end{bmatrix}$.

- 46.** Find, as a function of the parameter $k \in \mathbb{R}$, the rank of the matrices $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & k \\ 3 & k & 2k+1 \end{bmatrix}$, $\begin{bmatrix} k & 0 & -1 \\ -1 & 1 & k \\ 2k & 2 & 0 \end{bmatrix}$.

- 47.** Find $k \in \mathbb{R}$ such that the matrix $\begin{bmatrix} k & 3 & 2 \\ -1 & 2 & 1 \\ 2k+1 & k & 3 \end{bmatrix}$ is non-singular.

- 48.** Solve the following systems using matrix theory:

$$\text{a) } \begin{cases} 2x + 2y - z = 1 \\ x + y - z = 0 \\ 3x + 2y - 3z = 1 \end{cases} \quad \text{b) } \begin{cases} y + 2z = 3 \\ x + 2y + z = 5 \\ 2x - y - 3z = 0 \end{cases}$$

$$\text{c) } \begin{cases} x_1 + x_2 + x_3 - x_4 = 1 \\ -x_1 + x_2 + x_3 + x_4 = 1 \\ x_1 - x_2 + x_3 + x_4 = 1 \\ x_1 + x_2 + 3x_3 + x_4 = 1 \end{cases} \quad \text{d) } \begin{cases} x_1 + x_4 = 1 \\ x_2 + x_3 = 1 \\ x_1 + x_2 = 1 \\ x_3 + x_4 = 1 \\ x_1 + x_2 + x_3 + x_4 = 2 \end{cases}$$

- 49.** Using the augmented matrices of the following systems, solve them and classify them as to the existence of solutions, as a function of the real parameters a and b :

$$\text{a) } \begin{cases} x + 2y - z = 1 \\ x + 3y + az = a^2 \\ 2x + 3y - 3z = 3 \end{cases} \quad \text{b) } \begin{cases} x + 2y + z = 4 \\ 2x + 3y - 2z = 3 \\ -x - y + 3z = 1 \\ 3x + 5y + az = b \end{cases}$$

- 50.** Discuss the solutions of the system $\begin{cases} x - 3z = 3 \\ 2x + ky - 6z = 6 \\ x + y + kz = 1 + z \end{cases}$ as a function of the real parameter k .

- 51.** Discuss the solutions of the system $AX = B$ as a function of the real parameters a and b , where $A = \begin{bmatrix} 1 & a & ab \\ 2 & 0 & ab \\ 3 & a & ab \end{bmatrix}$ and $B = \begin{bmatrix} ab \\ 3ab \\ 5ab + a \end{bmatrix}$.

- 52.** Compute the inverse of each of the following matrices: a) $A = \begin{bmatrix} 2 & 1 & -1 \\ 2 & 1 & 0 \\ -8 & 0 & 0 \end{bmatrix}$ b) $B =$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix} \quad \text{c) } \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

- 53.** Consider the system $\begin{cases} x + 2y - z = 1 \\ x + 3y = 0 \\ 2x + 3y - 3z = 3 \end{cases}$. Solve it, given that $(x, y, z) = (3, -1, 0)$ is a solution.

- 54.** Consider the system $\begin{cases} x + y + z = 2 \\ 2x - y + z = 1 \\ 4x + y + 3z = 5 \end{cases}$. Solve it, given that $(x, y, z) = (1, 1, 0)$ is a solution.

55. Find which of the following matrices are skew-symmetric, symmetric, Hermitian or skew-Hermitian:

$$\begin{aligned} \text{a)} \begin{bmatrix} 1 & i & 1-i \\ -i & 0 & 1 \\ 1+i & 1 & 0 \end{bmatrix} \quad \text{b)} \begin{bmatrix} i & i & 1-i \\ i & 0 & 1 \\ -1-i & -1 & 0 \end{bmatrix} \quad \text{c)} \begin{bmatrix} 0 & 2 & 1 \\ 2 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} \\ \text{d)} \begin{bmatrix} 0 & 2 & 1 \\ -2 & 2 & -1 \\ -1 & 1 & 0 \end{bmatrix} \quad \text{e)} \begin{bmatrix} 0 & i & 1-i \\ i & 0 & 1 \\ -1-i & -1 & 0 \end{bmatrix} \quad \text{f)} \begin{bmatrix} 0 & 2 & 1 \\ 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \end{aligned}$$

56. Let A be a square matrix over a field $\mathbb{K}$. Show that:

- a) $(\lambda A)^T = \lambda A^T$, $\lambda \in \mathbb{K}$;
- b) $(A^p)^T = (A^T)^p$, $p \in \mathbb{N}_0$;
- c) $A + A^T$ and AA^T are symmetric matrices;
- d) $A - A^T$ is a skew-symmetric matrix.

57. Use the previous problem to:

- a) show that every square matrix can be written as the sum of a symmetric matrix and a skew-symmetric matrix (hint: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$);

- b) write the matrix $A = \begin{bmatrix} 3 & -4 & 7 \\ 3 & 1 & 9 \\ 8 & 6 & 9 \end{bmatrix}$ as the sum of a symmetric matrix and a skew-symmetric one.

58. Let A be a square matrix over the field of complex numbers. Show that: a) $A + A^*$ and AA^* are Hermitian matrices; b) $A - A^*$ is a skew-Hermitian matrix.

59. Use the previous problem to show that every square matrix can be written as the sum of a Hermitian matrix and a skew-Hermitian matrix. Write the matrix $A = \begin{bmatrix} 2+i & 3i \\ 5i & 4 \end{bmatrix}$ as the sum of a Hermitian matrix and a skew-Hermitian one.

60. Let $A^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$ be the inverse of the coefficient matrix of a system of equations written in matrix form $AX = B$.

- a) What can you conclude about the nature of the system?
- b) And about the linear dependence of the lines (rows and columns) of the matrix A ?
- c) Compute the solution matrix, X , given that $B^T = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$.
- d) Write the system $AX = B$ as a system of equations.

61. Solve the system $\begin{cases} x + z = 1 \\ x + y + z = 2 \\ 2x + 2y + z = -1 \end{cases}$ using the inverse of the coefficient matrix (solving explicitly for the unknowns).

62. Solve the matrix equation $AB^TX + (A + X^TB)^T = 0$, where $A = \begin{bmatrix} -2 & 2 & 0 \\ 0 & -2 & -1 \\ 2 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & -1 \\ 0 & -1 & 1 \\ 2 & -1 & 0 \end{bmatrix}$.

63. Consider the matrices $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 1 & 1 \\ 4 & 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, $C = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

a) Compute A^{-1} .

b) Solve the following systems of equations: i) $AX = B$ ii) $A^2X = C$ iii) $A^{10}X = D$ iv)

$$\begin{cases} 2x + y + 3z = 2 \\ y + z = -1 \\ 4x + 3z = 0 \end{cases} \quad \text{v) } \begin{cases} x + y - 1 = 0 \\ x + 3y + 2z = 0 \\ 3y + 4z + 2 = 0 \end{cases}$$

64. A power with a negative integer exponent of a non-singular matrix is defined as $A^{-n} = (A^{-1})^n =$

$(A^n)^{-1}$, $n \in \mathbb{N}$. Given the matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, compute A^{-2} .

65. A square matrix A is said to be orthogonal if $A^T A = A A^T = I$, that is, $A^T = A^{-1}$.

a) Show that $A = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$ is orthogonal.

b) Given that A and B are orthogonal matrices, show that $X = B - I$ is a solution of the matrix equation $A^T X^T B + A^{-1} B = A^T$.

c) Show that, if D is an orthogonal matrix, then $(D^{-1} B D)^T = D^{-1} B^T D$.

d) Show that the product of two orthogonal matrices is an orthogonal matrix.